Abhay Rajesh Patle

Nagpur, Maharashtra, India | +91 9356867020 | abhaypatle99@gmail.com | LinkedIn | GitHub

SUMMARY

Cloud & AI Engineer building event-driven AWS serverless architectures, production-grade FastAPI microservices, and AI-enabled systems. Architects secure, automated cloud workflows with AWS Lambda, Bedrock, Boto3, and Docker, applying least-privilege and zero-trust principles.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (Lambda, Bedrock, IAM, EC2, S3, VPC, RDS, DynamoDB, API Gateway, CloudWatch), Boto3, Serverless, Cloud Security, Least-Privilege Access
Backend & APIs: Python, FastAPI, Asynchronous REST APIs, JWT Authentication, Rate Limiting, Node.js, Express.js
DevOps, DB & Tools: Docker, Linux Administration, Git/GitHub, CI/CD, PostgreSQL, MongoDB, Pytest, Streamlit
AI & Data: PyTorch, XGBoost, Scikit-Learn, SHAP (Explainable AI), Computer Vision, Generative AI, RAG

CORE PROJECTS

IAM Guard AI Auditor v2.0 | *AWS Lambda, Amazon Bedrock, IAM, Python, Boto3, Serverless*

- Built an event-driven serverless auditor on AWS Lambda that analyzes IAM JSON policies in real time.
- Integrated Amazon Bedrock foundation models to compute 0–100 security scores and flag over-privileged and wildcard policies.
- Automated least-privilege remediation templates and side-by-side policy diffs, enforced with zero-trust guardrails.

Sahara – AI-Assisted Mental Healthcare & Wellness Platform | *React, Node.js, Express.js, MongoDB, FastAPI, Generative AI, JWT, Docker, Pytest*

- Architected a full-stack platform with a decoupled FastAPI microservice handling LLM-based conversational triage and risk classification.
- Secured endpoints with JWT refresh-token rotation, bcrypt hashing, CORS, and rate limiting; validated with Pytest.

Nagpur Pulse – AI Traffic Risk & Dynamic Police Deployment Support | *Python, XGBoost, SHAP, Scikit-Learn, FastAPI*

- Built async FastAPI REST endpoints serving real-time risk scores with under 50 ms latency.
- Led the ML pipeline for multivariate predictive modeling, using TreeSHAP explainability to support operational dispatch decisions.
- Finalist / 2nd Prize, Viksit Nagpur Hackathon 2026.

EDUCATION

B.Tech, Computer Science & Engineering | Tulsiramji Gaikwad-Patil College of Engineering & Technology (TGPCET), Nagpur | 2023–2027 | SGPA: 8.3/10

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Academy:** Cloud Foundations; Cloud Architecting
- **Viksit Nagpur Hackathon 2026:** Finalist / 2nd Prize (Nagpur Pulse)
- **Community:** Attendee/Contributor, AWS Student Community Day Nagpur and Google I/O Extended